

Assignment3:Basic Data Processing

Andre King

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Setup Workspace

```
#install.packages("tinytex")
tinytex::install_tinytex(force = TRUE)
```

```
## tlmgr --repository http://www.preining.info/tlgpg/ install tlgpg
```

```
## tlmgr option repository 'https://ctan.mirrors.hoobly.com/systems/texlive/tlnet'
```

```
## tlmgr update --list
```

```
#install.packages("tidyverse")
#install.packages("haven") # run once
library(haven)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.2      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dplyr)
library(tinytex)
getwd()
```

```
## [1] "/Users/senorking/Desktop/Fall2025/Data333"
```

```
setwd("/Users/senorking/Desktop/Fall2025/Data333/")
```

Import Data and create subset

Create a subset of the NHANES August 2021-August 2023 Demographics Data by keeping only these four columns – SEQN, SDDSRVYR, RIAGENDR, and RIDAGEYR. The data is imported then and new data is created using the subset of a few columns.

```
nhanes <- read_xpt("DEMO_L.xpt")
summary(nhanes)
```

```
##          SEQN          SDDSRVYR          RIDSTATR          RIAGENDR          RIDAGEYR
## Min.      :130378    Min.       :12    Min.       :1.000    Min.       :1.000    Min.       : 0.00
## 1st Qu.:133361    1st Qu.:12    1st Qu.:1.000    1st Qu.:1.000    1st Qu.:13.00
## Median :136344    Median :12    Median :2.000    Median :2.000    Median :37.00
## Mean     :136344    Mean     :12    Mean     :1.742    Mean     :1.533    Mean     :38.32
## 3rd Qu.:139327    3rd Qu.:12    3rd Qu.:2.000    3rd Qu.:2.000    3rd Qu.:62.00
## Max.     :142310    Max.      :12    Max.      :2.000    Max.      :2.000    Max.      :80.00
##
##          RIDAGEMN          RIDRETH1          RIDRETH3          RIDEXMON          RIDEXAGM
## Min.       : 0.00    Min.       :1.000    Min.       :1.000    Min.       :1.00    Min.       : 0.0
## 1st Qu.: 6.00    1st Qu.:3.000    1st Qu.:3.000    1st Qu.:1.00    1st Qu.: 66.0
## Median :11.00    Median :3.000    Median :3.000    Median :2.00    Median :122.0
## Mean     :11.63    Mean     :3.105    Mean     :3.321    Mean     :1.52    Mean     :121.9
## 3rd Qu.:17.00    3rd Qu.:4.000    3rd Qu.:4.000    3rd Qu.:2.00    3rd Qu.:179.5
## Max.     :24.00    Max.      :5.000    Max.      :7.000    Max.      :2.00    Max.      :239.0
## NA's      :11556                    NA's      :3073    NA's      :9146
##
##          DMMQILIZ          DMDBORN4          DMDYRUSR          DMDEDUC2
## Min.       :1.000    Min.       :1.000    Min.       : 1.000    Min.       :1.000
## 1st Qu.:2.000    1st Qu.:1.000    1st Qu.: 3.000    1st Qu.:3.000
## Median :2.000    Median :1.000    Median : 6.000    Median :4.000
## Mean     :1.917    Mean     :1.157    Mean     : 7.335    Mean     :3.805
## 3rd Qu.:2.000    3rd Qu.:1.000    3rd Qu.: 6.000    3rd Qu.:5.000
## Max.     :7.000    Max.      :2.000    Max.     :99.000    Max.      :9.000
## NA's      :3632    NA's      :19    NA's     :10058    NA's      :4139
##
##          DMDMARTZ          RIDEXPRG          DMDHHSIZ          DMDHRGND
## Min.       : 1.000    Min.       :1.000    Min.       :1.000    Min.       :1.000
## 1st Qu.: 1.000    1st Qu.:2.000    1st Qu.:2.000    1st Qu.:1.000
## Median : 1.000    Median :2.000    Median :3.000    Median :2.000
## Mean     : 1.778    Mean     :2.237    Mean     :3.243    Mean     :1.564
## 3rd Qu.: 2.000    3rd Qu.:3.000    3rd Qu.:4.000    3rd Qu.:2.000
## Max.     :99.000    Max.      :3.000    Max.      :7.000    Max.      :2.000
## NA's      :4141    NA's     :10430                    NA's      :7818
##
##          DMDHRAGZ          DMDHREDZ          DMDHRMAZ          DMDHSEDZ
## Min.       :1.00    Min.       :1.000    Min.       :1.000    Min.       :1.000
## 1st Qu.:2.00    1st Qu.:2.000    1st Qu.:1.000    1st Qu.:2.000
## Median :2.00    Median :2.000    Median :1.000    Median :2.000
## Mean     :2.54    Mean     :2.171    Mean     :1.381    Mean     :2.275
## 3rd Qu.:3.00    3rd Qu.:3.000    3rd Qu.:2.000    3rd Qu.:3.000
## Max.     :4.00    Max.      :3.000    Max.      :3.000    Max.      :3.000
## NA's      :7809    NA's      :8187    NA's      :7913    NA's      :9806
##
##          WTINT2YR          WTMEC2YR          SDMVSTRA          SDMVPSU
## Min.       : 4584    Min.       : 0    Min.       :173.0    Min.       :1.000
## 1st Qu.:14332    1st Qu.: 0    1st Qu.:176.0    1st Qu.:1.000
## Median :21670    Median :21718    Median :180.0    Median :1.000
## Mean     :27404    Mean       :27404    Mean     :179.9    Mean     :1.492
```

```
## 3rd Qu.: 33831 3rd Qu.: 38341 3rd Qu.:184.0 3rd Qu.:2.000
## Max. :170968 Max. :227108 Max. :187.0 Max. :2.000
##
## INDFMPIR
## Min. :0.000
## 1st Qu.:1.180
## Median :2.500
## Mean :2.708
## 3rd Qu.:4.500
## Max. :5.000
## NA's :2041
```

```
nh_subset <- nhanes[, c("SEQN", "SDDSRVYR", "RIAGENDR", "RIDAGEYR")]
view(nh_subset)
```

Rename the columns

In the demographics subset data, rename the column of “SDDSRVYR” as “cycle”, the column of “RIAGENDR” as “gender”, and the column of “RIDAGEYR” as “age”. Renaming this way we need to know the order of the columns so we do not rename the wrong column.

Filter Data

Exclude observations (i.e., study participants) who are younger than 18 years old (i.e., age < 18) from the demo_sub dataset. Count and report the number of remaining observations (i.e., study participants aged 18 and older).

```
nrow(nh_subset) # 11933 rows before
```

```
## [1] 11933
```

```
nh_subset <- nh_subset %>%
  filter(age >= 18)
nrow(nh_subset) #8153 participants aged 18 and older
```

```
## [1] 8153
```

Before the data was filtered there were 11933 observations, after filtering out participants aged 18 and younger there is 8153 participants aged 18 and older remaining.

Check if SEQN is a unique identifier (no duplicates)

```
table(count(nh_subset, SEQN)$n)
```

```
##
## 1
## 8153
```

```
sum(duplicated(nh_subset$SEQN))
```

```
## [1] 0
```

There are 0 duplicates meaning SEQN is a unique identifier so it can be used to join/merge data

Merge the data

In order to merge data, all data must be imported

```
dpq <- read_xpt("DPQ_L.xpt")
view(dpq)
sum(duplicated(dpq$SEQN))
```

```
## [1] 0
```

```
seqn.merge <- left_join(nh_subset, dpq, join_by(SEQN))
seqn.merge
```

```
## # A tibble: 8,153 x 14
##   SEQN cycle gender  age DPQ010 DPQ020 DPQ030 DPQ040 DPQ050 DPQ060 DPQ070
##   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 130378    12     1   43    NA    NA    NA    NA    NA    NA    NA
## 2 130379    12     1   66     0     0     1     0     0     0     0
## 3 130380    12     2   44     0     0     1     1     0     0     0
## 4 130384    12     1   43    NA    NA    NA    NA    NA    NA    NA
## 5 130385    12     2   65    NA    NA    NA    NA    NA    NA    NA
## 6 130386    12     1   34     0     0     0     0     0     1     0
## 7 130387    12     2   68     0     0     0     0     0     0     0
## 8 130388    12     2   27    NA    NA    NA    NA    NA    NA    NA
## 9 130389    12     1   59     0     0     0     0     0     0     0
## 10 130390    12     2   31    NA    NA    NA    NA    NA    NA    NA
## # i 8,143 more rows
## # i 3 more variables: DPQ080 <dbl>, DPQ090 <dbl>, DPQ100 <dbl>
```

```
nrow(seqn.merge)
```

```
## [1] 8153
```

In the merged data “seqn_merge” there are 8153 observations